

Sayana S Nair

☎ +91-9863329260

✉ cb.ps.i5das22144@cb.students.amrita.edu

in linkedin.com/in/sayanair

Objective

Motivated Data Science student with a strong foundation in Python, mathematics, statistics, and machine learning. Proficient in data analysis, statistical modeling, and data visualization. Passionate about deriving actionable insights from complex datasets and contributing to innovative solutions in data-driven decision-making. Eager to leverage technical expertise and analytical skills during an internship to gain hands-on industry experience.

Education

Integrated MSc in Data Science

Amrita Vishwa Vidyapeetham, Coimbatore 2022 – Present
Pursuing Final Year | CGPA: 8.07

Higher Secondary Education (12th Grade)

Vivekananda Kendra Vidyalaya, Nirjuli 2018 – 2020
Science / Physics, Chemistry, Biology, Mathematics | CBSE | 86

Secondary Education (10th Grade)

Vivekananda Kendra Vidyalaya, Nirjuli 2016 – 2018
CBSE | 88

Skills

Technical Skills

- Programming Languages: Python, R, MATLAB
- Machine Learning Frameworks: Scikit-learn
- Data Visualization: Matplotlib, Seaborn
- Tools: Excel, Jupyter Notebook, Git, Pandas, NumPy

Core Competencies

- Data Structures and Algorithms
- Statistical Analysis (Descriptive and Inferential)
- Linear Algebra and Mathematical Modeling
- Natural Language Processing (NLP) Fundamentals

Projects

Data Visualization on Natural Disasters

- Analyzed historical natural disaster datasets to identify trends and patterns.

- Used Tableau and Matplotlib to create interactive dashboards visualizing disaster frequency, impact, and geographic distribution.
- Insights provided actionable recommendations for disaster management strategies.

MQTT IoT Dataset - Intrusion Detection

- Utilized machine learning tools to analyze the MQTT IoT Dataset and derive actionable insights for intrusion detection.
- Preprocessed and analyzed dataset containing both normal and anomalous network traffic patterns using techniques like feature extraction, data cleaning, and normalization.
- Applied machine learning algorithms such as Random Forest, Support Vector Machines (SVM), and Neural Networks to classify and detect potential intrusions.

Medical Image Processing with Basic Graphs

- Currently working on a project involving medical image processing.
- Applied graph-based algorithms to analyze and interpret medical datasets.
- Aiming to derive meaningful insights for healthcare applications.

Application of Heat Transfer in Biomedical Engineering

- Gained foundational knowledge in simulating heat transfer in biological tissues during hyperthermia treatments under the guidance of a professor from the Mathematics Department.
- Explored thermal diffusion equations to predict temperature distributions in biological tissues using MATLAB and Python.
- Developed insights into optimizing treatment parameters for hyperthermia therapy, ensuring minimal damage to healthy tissues while targeting cancerous cells.
- Demonstrated proficiency in interdisciplinary research by combining principles of mathematics, physics, and biomedical engineering.

Certifications and Training

Natural Language Processing Fundamentals

Infosys Springboard February 03, 2024
Explored NLP concepts like tokenization, stemming, lemmatization, and sentiment analysis.

Machine Learning with MATLAB

Explore Machine Learning with MATLAB January 31, 2024
Learned to implement machine learning algorithms and analyze datasets using MATLAB.

Career Essentials in Data Analytics

LinkedIn and Microsoft *[June 2024]*
Developed foundational skills in analysis tools.

Infosys Pragati Cohort 4

Ongoing Training Program *(Enrolled)*
Infosys Springboard's "Pragati: Path to Future Cohort 4" is a 3.5-month skilling program for women, focusing on self-growth, professional advancement, career development, and leadership skills.

Interests

Passionate about data-driven projects, particularly in leveraging machine learning and AI for real-world applications. Keen interest in exploring advanced data structures, algorithms, and their applications in optimizing computational efficiency and scalability.

Additional Achievements

Scholarships

- National Means cum Merit Scholarship, Arunachal Pradesh 2016

Extracurricular Activities

- Participated in Creative Writing (Hindi and English), Short Film Competitions, Wall Magazine, Poster Making, and Science Exhibitions.

Languages

- English and Hindi - Professional Working Proficiency
- Malayalam and Sanskrit - Native / Bilingual Proficiency

Personal Details

Date of Birth: 26th September 2002

Hobbies / Interests: Dance, Poster Making, Story Writing

Contact Address

House No. 626, D – Sector, Naharlagun, District – Papum Pare, State – Arunachal Pradesh, Pincode - 791110